Chapter 4. Focal points: every manifold has a Morse function

Milnor, "Morse Theory", §6, rewritten for a first-year graduate student who wants to know what the theorem is for. Every proof is complete; Sard's theorem, the normal bundle, the index of a nondegenerate zero and the Poincaré–Hopf theorem are quoted from the companion volume ("Topology from the Differentiable Viewpoint", rewritten), and the Whitney embedding theorem from the literature.

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0. Echoes in an oval room

Stand in a room whose wall, seen from above, is an ellipse, and clap once. Some of the sound strikes the wall head-on and comes straight back to you. How many such echoes do you hear, and when?

A sound ray from your position p returns to p after one reflection at a point q of the wall exactly when it meets the wall at right angles: the reflected ray is the mirror image of the incoming one in the normal line, and it points back along the incoming ray only if that ray is the normal. So the echoes correspond to the normals to the wall through p, and these are the critical points of the squared distance

$$L_p(q) = |p - q|^2, \qquad q \text{ on the wall},$$

because the derivative of L_p along the wall vanishes exactly when $p - q$ is perpendicular to the wall. The delay of an echo is $2|p - q|$ divided by the speed of sound.

The answer depends on where you stand, through a curve you cannot see: the locus of the centres of curvature of the wall, its *evolute*. Inside the evolute you hear four echoes, outside it two. As you walk across it, two of the echoes slide together in time, merge and vanish; on the evolute itself L_p has a degenerate critical point.

This chapter proves the general statement behind this, for any submanifold M of a Euclidean space. L_p has a degenerate critical point exactly when p is a *focal point* of M, a point where infinitely near normals of M meet (Lemma 6.5). Along each normal line the focal points sit at the principal radii of curvature (Lemma 6.3), and by Sard's theorem they form a set of measure zero. Hence L_p is a Morse function for almost every p (Theorem 6.6). Since every manifold embeds in some $\mathbb{R}^n$ as a closed subset, every manifold carries a Morse function with compact sublevel sets (Corollary 6.7), and the theory of Chapters 2 and 3 applies to every manifold. Finally, the index of L_p at q is the number of focal points between q and p (Lemma 6.9): the echoes that come back from beyond a centre of curvature are those of index 1. Section 9 does the numbers for an oval room 20 m long and 12 m wide.

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1. Normals and focal points (Milnor p. 32)

So far we have studied functions with only nondegenerate critical points, without knowing that any exist. Milnor constructs them on every manifold embedded in Euclidean space.

Setting. Throughout, $M \subset \mathbb{R}^n$ is a smooth submanifold of dimension $k < n$ (Milnor: "differentiably embedded"; M need not be compact or closed). Every point of M has a neighbourhood $x(U)$ in M, where $U \subset \mathbb{R}^k$ is open and $x : U \to \mathbb{R}^n$ is a *parametrization*: smooth, with injective

derivative, and a homeomorphism onto $x(U)$. Its inverse gives coordinates $u = (u^1, \dots, u^k)$, and TM_q is the image of dx at q. We cite the companion volume by chapter and number.

Definition (Milnor). The *normal bundle* is $N = \{(q, v) \in M \times \mathbb{R}^n : v \perp TM_q\}$, and the *endpoint map* is $E : N \to \mathbb{R}^n$, $E(q, v) = q + v$.

Fact A (the normal bundle). N is a smooth n-dimensional submanifold of $\mathbb{R}^{2n}$. Concretely, let $x : U \to M$ be a parametrization and $w_1, \dots, w_{n-k} : U \to \mathbb{R}^n$ smooth maps such that $w_1(u), \dots, w_{n-k}(u)$ is an orthonormal basis of $(TM_{x(u)})^\perp$ for each u. Then

$$\Psi(u, t) = \Big(x(u),\, \sum_{\alpha=1}^{n-k} t^\alpha w_\alpha(u)\Big), \qquad (u, t) \in U \times \mathbb{R}^{n-k},$$

is a parametrization of N onto the open subset $N \cap (x(U) \times \mathbb{R}^n)$. Such frames exist near every point of M: apply Gram–Schmidt to $\partial_1 x, \dots, \partial_k x$ followed by $n - k$ standard basis vectors completing them to a basis at one point (they then do so nearby). This is proved in Chapter 6 of the companion volume (*Topology from the Differentiable Viewpoint*, rewritten), in the proof of Proposition 4.1; see also its Chapter 8, Problem 11. That proof assumes M compact but is local. (In brief: Ψ is smooth and injective; its derivative sends $\partial/\partial u^i$ and $\partial/\partial t^\beta$ to $(\partial_i x, *)$ and $(0, w_\beta)$, which are independent; and its inverse $(q, v) \mapsto (x^{-1}(q), (v \cdot w_\alpha(x^{-1}(q)))_\alpha)$ is continuous.)

E is smooth, as the restriction of the linear map $(q, v) \mapsto q + v$ of $\mathbb{R}^{2n}$.

Definition (Milnor). A point $e \in \mathbb{R}^n$ is a *focal point of (M, q) with multiplicity μ* if $e = q + v$ with $(q, v) \in N$ and the derivative $dE_{(q,v)} : TN_{(q,v)} \to \mathbb{R}^n$ has a kernel of dimension $\mu > 0$. It is a *focal point of M* if it is a focal point of (M, q) for some $q \in M$.

Both spaces have dimension n, so (q, v) is then a critical point of E. Intuitively a focal point is where nearby normals meet: if dE kills a tangent vector to N at (q, v), then the normal lines through nearby points of M, followed for the right length, arrive at e to first order (Milnor's figure on p. 32).

Example. For the sphere $M = S^{n-1}(r)$ of radius r about 0, the normal space at q is spanned by q, and in the parametrization $\Psi(u, s) = (x(u), s\, x(u)/r)$ we get $E(\Psi(u, s)) = (1 + s/r)\, x(u)$, with partial derivatives $(1 + s/r)\, \partial_i x$ and x/r. These are dependent exactly when $s = -r$, and then the kernel has dimension $n - 1$. So the centre is a focal point of (M, q) for every q, of multiplicity $n - 1$, and there are no others: all normals of a sphere meet at its centre.

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2. Sard's theorem and the measure of the focal set (Milnor p. 33)

A set $Z \subset \mathbb{R}^n$ has *measure zero* if for each $\eta > 0$ it can be covered by countably many cubes of total volume less than η. A countable union of such sets has measure zero, and a set of measure zero contains no ball, so its complement is dense. A *critical point* of a C^1 map between manifolds of the same dimension is a point where its derivative is singular.

Theorem 6.1 (Sard). *(checked by machine: V4.3, in a narrower form)*[1] If M_1 and M_2 are manifolds with countable bases, of the same dimension, and $f : M_1 \to M_2$ is of class C^1, then the image of

[1] Checked by machine: the result has been verified in the Lean proof assistant. The numbers are those of the verified edition and of the comparison, which give the checked statement in full. *As stated*: the same statement. *Restated*: an equivalent statement in other terms. *With its hypotheses made explicit*: a hypothesis the book uses silently is stated. *In*

the set of critical points has measure zero in M_2 (its trace in each chart is carried to a set of measure zero).

Milnor does not prove it and cites de Rham, *Variétés différentiables*, p. 10. We apply it only to smooth maps into $\mathbb{R}^n$, and that case is proved in Chapter 3 of the companion volume: Theorem 1.1 there is Sard's theorem for smooth maps between open subsets of vector spaces, and the proof of its Theorem 2.1 passes to a manifold domain by covering it with countably many charts φ_i, the critical values of f being the union of those of the maps $f \circ \varphi_i^{-1}$.

> **How much smoothness.** "Of class C^1" suffices because the dimensions are equal. From higher-dimensional manifolds Sard's theorem needs more derivatives, and Whitney's C^1 example shows that they are necessary (companion volume, Chapter 3). Every map below is smooth.

Corollary 6.2. *(checked by machine: V4.4, with its hypotheses made explicit)* For almost all $x \in \mathbb{R}^n$ (all but a set of measure zero), x is not a focal point of M.

Proof. N is an n-manifold (Fact A), with a countable basis since it is a subset of $\mathbb{R}^{2n}$, and $E : N \to \mathbb{R}^n$ is smooth. By definition, x is a focal point of M if and only if $x = E(q, v)$ for a point $(q, v) \in N$ at which dE is singular, that is, if and only if x is a critical value of E. By Theorem 6.1 these form a set of measure zero. □

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3. The second fundamental form (Milnor pp. 33–34)

Fix a parametrization $x(u) = (x_1(u), \dots, x_n(u))$ of a neighbourhood of q in M, write $\partial_i = \partial/\partial u^i$, and let u_q be the coordinates of q. (Milnor writes vectors with arrows; we drop them.)

Definition (Milnor). The *first fundamental form* is the symmetric matrix of functions $g_{ij} = \partial_i x \cdot \partial_j x$; it is positive definite because the $\partial_i x$ are independent. The vector $\partial_i \partial_j x$ at a point of M is the sum of a vector tangent to M and a vector normal to M; the normal part is denoted ℓ_{ij}, and the matrix (ℓ_{ij}) of vector-valued functions, symmetric in i, j, is the *second fundamental form*. For a unit normal vector v at q, the symmetric matrix

$$(v \cdot \partial_i \partial_j x) = (v \cdot \ell_{ij})$$

is the *second fundamental form of M at q in the direction v*. The two expressions agree because v is orthogonal to the tangent part.

It simplifies matters to have $g_{ij}(u_q) = \delta_{ij}$. A linear change of coordinates achieves this: if P is invertible with $P^{\mathsf{T}} g(u_q) P = I$ (for instance $P = g(u_q)^{-1/2}$), then $\tilde{x}(\tilde{u}) = x(u_q + P\tilde{u})$ has $\tilde{g}(0) = I$.

Definition (Milnor). With $g_{ij}(u_q) = \delta_{ij}$, the eigenvalues $K_1, \dots, K_k$ of the symmetric $k \times k$ matrix $(v \cdot \ell_{ij})$ at q are the *principal curvatures* of M at q in the normal direction v, and the reciprocals K_i^{-1} of the nonzero ones are the *principal radii of curvature*. (Milnor writes $K_1, \dots, K_n$; there are k of them.) If the matrix is singular some K_i vanish, and the corresponding radii are not defined.

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a stronger form: the checked statement says more than the book's. *In a narrower form*: it covers less, a special setting or a weaker conclusion. *With a sign corrected*: the book's statement has a slip. *Not checked*: no part of the result has been verified. The section "Checked by machine" at the end of the chapter says which forms were checked, and covers the results that carry no such note.

Claim B. The principal curvatures, with multiplicities, do not depend on the coordinates with $g_{ij}(u_q) = \delta_{ij}$. In arbitrary coordinates they are the eigenvalues of $g^{-1}(v \cdot \ell)$ at q.

Proof. Let $\tilde{u}$ be other coordinates near q and $J = (\partial u^i / \partial \tilde{u}^a)$ at q. By the chain rule $\partial x / \partial \tilde{u}^a = \sum_i J_{ia}\, \partial_i x$, so $\tilde{g} = J^{\mathsf{T}} g J$, and

$$\frac{\partial^2 x}{\partial \tilde{u}^a \partial \tilde{u}^b} = \sum_{i,j} J_{ia} J_{jb}\, \partial_i \partial_j x + \sum_i \frac{\partial^2 u^i}{\partial \tilde{u}^a \partial \tilde{u}^b}\, \partial_i x.$$

The last sum is tangent, so $v \cdot \tilde{\ell} = J^{\mathsf{T}}(v \cdot \ell) J$. If $g = \tilde{g} = I$ at q then $J^{\mathsf{T}} J = I$, and $v \cdot \tilde{\ell} = J^{-1}(v \cdot \ell) J$ has the same eigenvalues. In general $\tilde{g}^{-1}(v \cdot \tilde{\ell}) = J^{-1} g^{-1} (v \cdot \ell) J$, which for $g = I$ is the matrix of the definition. □

Example. For the sphere of radius r and the inward normal $v = -q/r$: differentiating $x \cdot x = r^2$ twice gives $\partial_i \partial_j x \cdot x = -g_{ij}$, so $v \cdot \ell_{ij} = g_{ij}/r$. All principal curvatures equal $1/r$ and all radii equal r, in agreement with §1.

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4. Focal points along a normal line (Milnor pp. 34–35)

Let v be a unit normal vector at q and $\ell = \{q + tv : t \in \mathbb{R}\}$ the normal line.

Lemma 6.3. *(checked by machine: V4.7, in a stronger form)* The focal points of (M, q) on ℓ are exactly the points $q + K_i^{-1} v$ with $K_i \neq 0$, where $K_1, \dots, K_k$ are the principal curvatures in the direction v; the multiplicity of $q + K^{-1} v$ is the multiplicity of K as an eigenvalue. Hence there are at most k focal points of (M, q) on ℓ, counted with multiplicity.

Proof. Choose x and a normal frame $w_1, \dots, w_{n-k}$ near q (Fact A) and use the coordinates (u, t) on N given by Ψ. Then $e(u,t) := E(\Psi(u,t)) = x(u) + \sum_\alpha t^\alpha w_\alpha(u)$, with

$$\frac{\partial e}{\partial u^i} = \partial_i x + \sum_\alpha t^\alpha\, \partial_i w_\alpha, \qquad \frac{\partial e}{\partial t^\beta} = w_\beta.$$

Since $d\Psi$ is an isomorphism onto the tangent space of N, the kernel of dE at $\Psi(u,t)$ has the dimension of the kernel of the $n \times n$ Jacobian matrix of e. Take the inner products of these n vectors with the basis $\partial_1 x, \dots, \partial_k x, w_1, \dots, w_{n-k}$ of $\mathbb{R}^n$. This multiplies the Jacobian by an invertible matrix and so does not change the dimension of the kernel. Since $w_\alpha \cdot \partial_j x = 0$ and $w_\alpha \cdot w_\beta = \delta_{\alpha\beta}$, with rows indexed by u^i then t^β, and columns by $\partial_j x$ then w_β, the result is

$$\begin{pmatrix} \partial_i x \cdot \partial_j x + \sum_\alpha t^\alpha\, \partial_i w_\alpha \cdot \partial_j x & \sum_\alpha t^\alpha\, \partial_i w_\alpha \cdot w_\beta \\ 0 & I \end{pmatrix}.$$

Differentiating $w_\alpha \cdot \partial_j x = 0$ gives $\partial_i w_\alpha \cdot \partial_j x = -w_\alpha \cdot \partial_i \partial_j x = -w_\alpha \cdot \ell_{ij}$, so the upper left block is $(g_{ij} - \sum_\alpha t^\alpha\, w_\alpha \cdot \ell_{ij})$. Subtracting multiples of the identity rows clears the upper right block, so the kernel of the whole matrix has the dimension of the kernel of the upper left block. At the point (q, tv) of N we have $\sum_\alpha t^\alpha w_\alpha = tv$, hence $\sum_\alpha t^\alpha w_\alpha \cdot \ell_{ij} = t\, v \cdot \ell_{ij}$. We have proved:

Assertion 6.4. *(checked by machine: V4.9, in a stronger form)* $q + tv$ is a focal point of (M, q) with multiplicity μ if and only if the matrix

$$(g_{ij} - t\, v \cdot \ell_{ij}) \qquad (*)$$

at q is singular, with nullity μ.

Now take coordinates with $g_{ij}(u_q) = \delta_{ij}$ and let $L = (v \cdot \ell_{ij}(u_q))$. For $t = 0$ the matrix $(*)$ is I, so q itself is never a focal point of (M, q). For $t \neq 0$, $I - tL = -t(L - t^{-1}I)$ is singular if and only if $1/t$ is an eigenvalue of L, and its nullity is the dimension of the $1/t$-eigenspace, which for the symmetric matrix L is the multiplicity of $1/t$ as an eigenvalue. So the focal points on ℓ are the points $q + tv$ with $1/t = K_i$, with the stated multiplicities, which add up to at most k. □

> **Milnor's text skips three small points.** (1) The fields w_α, which he writes as functions of u, need Gram–Schmidt to exist near each point (Fact A), and in general they exist only locally, not along all of M: a Möbius band in $\mathbb{R}^3$ has no nowhere-zero normal field. (2) "Thus the rank is equal to the rank of the upper left hand block": the rank of the $n \times n$ matrix is that rank plus $n - k$; it is the nullities that agree, which is what the argument uses. (3) The case $t = 0$ must be set aside before dividing by t, and "multiplicity as eigenvalue" equals the nullity because $(v \cdot \ell_{ij})$ is symmetric.

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5. The squared distance function (Milnor pp. 35–36)

For $p \in \mathbb{R}^n$ let $L_p : M \to \mathbb{R}$, $L_p(q) = |q - p|^2$. In coordinates, $f(u) := L_p(x(u)) = x \cdot x - 2\,x \cdot p + p \cdot p$, so

$$\partial_i f = 2\,\partial_i x \cdot (x - p), \qquad \partial_i \partial_j f = 2(\partial_i x \cdot \partial_j x + \partial_i \partial_j x \cdot (x - p)).$$

The first formula shows that *q is a critical point of L_p if and only if $q - p$ is normal to M at q*, that is, $(q, p - q) \in N$. So the critical points of L_p are the first components of the points of $E^{-1}(p)$.

At a critical point $q \neq p$ write $p = q + tv$, where $t > 0$ is the distance from q to p and $v = (p - q)/t$ is a unit normal. Then $\partial_i \partial_j x \cdot (x - p) = -t\,v \cdot \partial_i \partial_j x = -t\,v \cdot \ell_{ij}$, so the Hessian is

$$\partial_i \partial_j f(u_q) = 2(g_{ij} - t\,v \cdot \ell_{ij}). \qquad (**)$$

If $p = q$, the Hessian is $2g_{ij}$, nondegenerate of index 0 (q is the minimum). Recall from Chapter 1 that at a critical point the Hessian is a well-defined symmetric bilinear form on TM_q, so its nullity and index do not depend on the coordinates.

Lemma 6.5. *(checked by machine: V4.11, with its hypotheses made explicit)* The point $q \in M$ is a degenerate critical point of L_p if and only if p is a focal point of (M, q). The nullity of q as a critical point equals the multiplicity of p as a focal point.

Proof. If q is not critical, then $p - q$ is not normal at q and p is not a focal point of (M, q). If q is critical and $q = p$, the Hessian $2g$ is nondegenerate and $p = q$ is not a focal point of (M, q) (Lemma 6.3, $t = 0$). If q is critical and $q \neq p$, the Hessian $(**)$ is twice the matrix $(*)$ of Assertion 6.4, so the nullities agree. □

Theorem 6.6. *(checked by machine: V4.12, in a stronger form)* For almost all $p \in \mathbb{R}^n$, the function $L_p : M \to \mathbb{R}$ has no degenerate critical points.

Proof. By Lemma 6.5, L_p has a degenerate critical point if and only if p is a focal point of (M, q) for some q, that is, a focal point of M. By Corollary 6.2 these p form a set of measure zero. □

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6. Every manifold carries a Morse function (Milnor pp. 36–37)

Here M is an abstract manifold (smooth, Hausdorff, with a countable basis), of dimension m. To use Theorem 6.6 we need it inside a Euclidean space.

Whitney embedding theorem (quoted). Every smooth m-manifold that is Hausdorff and has a countable basis admits a smooth embedding into $\mathbb{R}^{2m+1}$ whose image is a closed subset. (Milnor cites Whitney, *Geometric Integration Theory*, p. 113; a modern proof is in J. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Chapter 6.)

For compact M an easy version suffices. Cover M by charts $\varphi_i : U_i \to \mathbb{R}^m$, $i = 1, \dots, r$, and choose smooth $\rho_i : M \to [0,1]$ supported in U_i such that the interiors of the sets $\{\rho_i = 1\}$ cover M. Then $F = (\rho_1\varphi_1, \dots, \rho_r\varphi_r, \rho_1, \dots, \rho_r)$ is injective (if $F(a) = F(b)$ and $\rho_i(a) = 1$, then $\rho_i(b) = 1$, both points lie in U_i, and $\varphi_i(a) = \varphi_i(b)$) and an immersion (near a point with $\rho_i = 1$ the component $\rho_i\varphi_i = \varphi_i$ is a chart), hence an embedding, M being compact.

Corollary 6.7. *(checked by machine: V4.13, V4.17, in a narrower form)* On any manifold M there is a smooth function with no degenerate critical points for which each M^a is compact.

Proof. Embed M as a closed subset of $\mathbb{R}^{2m+1}$ and identify it with its image. By Theorem 6.6 choose p such that L_p has no degenerate critical points. Then $M^a = L_p^{-1}(-\infty, a] = M \cap \bar{B}(p, \sqrt{a})$ (empty for $a < 0$) is closed and bounded in $\mathbb{R}^{2m+1}$, hence compact. □

> **A hidden hypothesis.** "Any manifold" must be Hausdorff with a countable basis, as Whitney's theorem requires, and the corollary fails without it: on the long line every continuous real function is eventually constant, so some sublevel set contains a closed unbounded piece of the line and is not compact. The closedness of the embedding is what makes M^a compact; an embedding onto a non-closed set would not do.

Application 1 (Milnor). Every manifold has the homotopy type of a CW complex, with one cell of dimension λ for each critical point of index λ of the function of Corollary 6.7. *Proof.* Theorem 3.5 (Chapter 2) applies to that function. □

Application 2 (Milnor). Let M be compact and f a Morse function on M (one exists by Theorem 6.6). Then: (i) $\chi(M) = \sum_\lambda (-1)^\lambda C_\lambda$; (ii) for any Riemannian metric, the zeros of grad f are the critical points of f, and the index of grad f at a critical point of index λ is $(-1)^\lambda$; hence (iii) the vector field grad f has index sum $\chi(M)$, and so does every vector field on M with isolated zeros.

Proof. (i) is Theorem 5.2, equation (3), of Chapter 3. (ii) In coordinates grad f has components $X^i = \sum_j g^{ij}\partial_j f$, which vanish exactly where $df = 0$. At a critical point the terms containing $\partial_j f$ drop out of $\partial_k X^i$, so the Jacobian matrix of X there is $G^{-1}H$, with $G = (g_{ij})$ and H the Hessian. Its determinant $\det H / \det G$ is nonzero, with the sign of $\det H$, which is $(-1)^\lambda$ since H has λ negative and $m - \lambda$ positive eigenvalues. A nondegenerate zero has index equal to the sign of this determinant (companion volume, Chapter 6, Lemmas 3.1 and 3.2). (iii) Embed M in some $\mathbb{R}^N$. The index sum of a tangent vector field with isolated zeros does not depend on the field: companion volume, Chapter 6, Theorem 4.2 for nondegenerate zeros and the remark on degenerate zeros that follows it (Milnor cites Steenrod, *The Topology of Fibre Bundles*, 39.7). □

> **Milnor's text skips** the hypothesis in (i): "for any differentiable function f on M" must read "for any function without degenerate critical points". This application is also the identification that the companion volume (Chapter 6) quotes from this book: the Euler number defined there by vector fields equals the homological Euler characteristic.

7. Approximation by Morse functions (Milnor p. 37)

Corollary 6.8. *(checked by machine: V4.21, in a stronger form; V4.22, in a narrower form)* Let M be a manifold of dimension m, $f : M \to \mathbb{R}$ a bounded smooth function, $K \subset M$ compact and $k \geq 0$ an integer. Fix finitely many charts $\varphi_j : U_j \to \mathbb{R}^m$ and compact sets $K_j \subset U_j$ with $K = \bigcup_j K_j$. Then for every $\eta > 0$ there is a smooth $g : M \to \mathbb{R}$ without degenerate critical points such that $|g - f| < \eta$ on M, and every partial derivative of order between 1 and k of $(g - f) \circ \varphi_j^{-1}$ has absolute value less than η on $\varphi_j(K_j)$, for every j.

Milnor asks that on the compact set K the derivatives of g of order $i \leq k$ uniformly approximate those of f. Derivatives on a manifold make sense only in charts, so the charts must be fixed in advance; finitely many suffice because K is compact. (Milnor compares M. Morse, Trans. Amer. Math. Soc. 33 (1931), 71–91.)

Proof. Step 1: a bounded embedding whose first coordinate is f. Let $h' : M \to \mathbb{R}^{n-1}$ be an embedding (Whitney), $\beta(y) = y/\sqrt{1+|y|^2}$, a diffeomorphism of $\mathbb{R}^{n-1}$ onto the open unit ball, and $h = (f, \beta \circ h') : M \to \mathbb{R}^n$. Then h is smooth, injective and an immersion (because its second component is), and a homeomorphism onto its image (its inverse is $(\beta \circ h')^{-1}$ composed with the projection forgetting the first coordinate). So h is an embedding with $h_1 = f$, and $|h| \leq B := (\sup f^2 + 1)^{1/2}$.

Step 2: Milnor's "short computation". For $c > 0$ and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)$ put $p = (-c + \varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)$ and $g = (L_p \circ h - c^2)/(2c)$. Expanding $|h - p|^2 = (h_1 + c - \varepsilon_1)^2 + \sum_{i \geq 2} (h_i - \varepsilon_i)^2$,

$$g = f + \sum_{i=1}^{n} \frac{h_i^2}{2c} - \sum_{i=1}^{n} \frac{\varepsilon_i h_i}{c} + \sum_{i=1}^{n} \frac{\varepsilon_i^2}{2c} - \varepsilon_1 .$$

Step 3: estimates. The formula says $g - f = |h - \varepsilon|^2/(2c) - \varepsilon_1$, so $|g - f| \leq (B + |\varepsilon|)^2/(2c) + |\varepsilon|$. In the derivatives of order ≥ 1 the constants drop out; the functions $h_i \circ \varphi_j^{-1}$ and their squares are smooth on $\varphi_j(U_j)$, so their partial derivatives of orders 1 to k are bounded on the compact sets $\varphi_j(K_j)$ by one constant D. Hence each such derivative of $(g - f) \circ \varphi_j^{-1}$ is at most $nD/(2c) + n|\varepsilon|D/c$. Choose $c > \max\{(B+1)^2/\eta,\ 3nD/(2\eta),\ 1\}$; then every ε with $|\varepsilon| < \min\{1, \eta/2\}$ gives both estimates with bound η.

Step 4: nondegeneracy. By Theorem 6.6 for the submanifold $h(M) \subset \mathbb{R}^n$, the points p for which L_p has a degenerate critical point on $h(M)$ form a set of measure zero, which contains no ball. So some ε in the allowed range makes $L_p \circ h$ a Morse function, and then so is g, which has the same critical points and Hessians multiplied by $1/(2c) > 0$. □

> **Why "bounded".** Boundedness of f is what makes h bounded, and hence the error terms $\sum h_i^2/2c$ uniformly small. Without it the statement is still true but needs another argument (density of Morse functions in the strong topology; M. Hirsch, *Differential Topology*, Chapter 6, §1).

8. The index of L_p (Milnor pp. 37–38)

Lemma 6.9 (index theorem for L_p). *(checked by machine: V4.23, restated)* The index of L_p at a nondegenerate critical point $q \in M$ equals the number of focal points of (M, q) on the segment from q to p, each counted with its multiplicity.

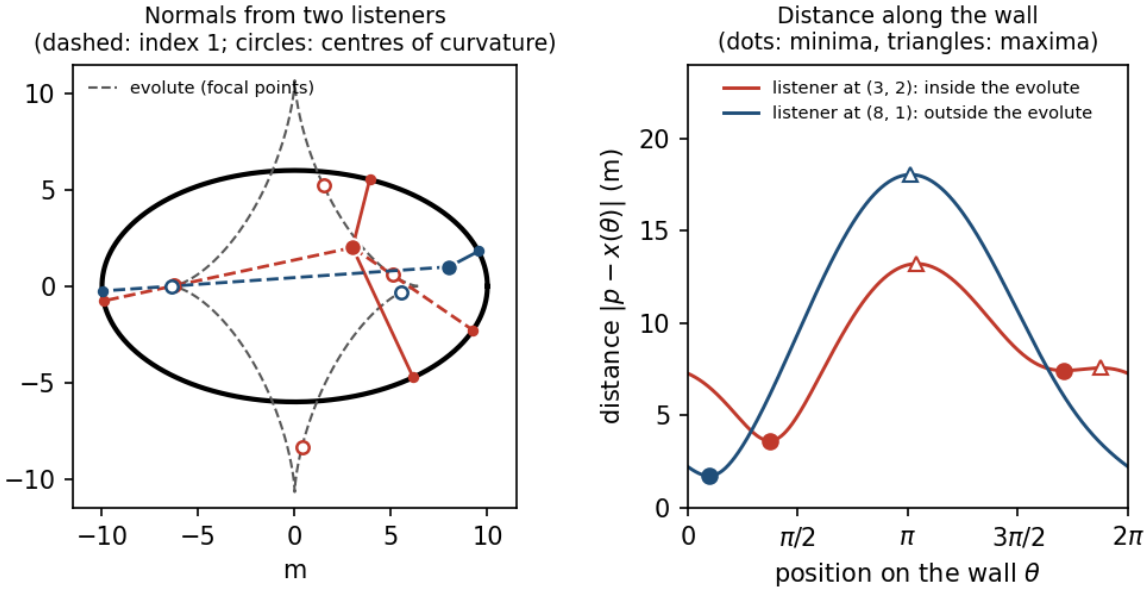
Proof. If $p = q$, the index is 0 and the segment $\{q\}$ contains no focal point of (M, q) (Lemma 6.3, $t = 0$). Otherwise write $p = q + tv$ with $t > 0$ as in §5 and take coordinates with $g_{ij}(u_q) = \delta_{ij}$. The index of the Hessian $(**)$ is the number of negative eigenvalues of $I - tL$, $L = (v \cdot \ell_{ij})$: by the spectral theorem a symmetric matrix is negative definite on the span of its eigenvectors with negative eigenvalues and nonnegative on a complement. The eigenvalues of $I - tL$ are $1 - tK_i$, and $1 - tK_i < 0$ if and only if $K_i > 1/t$, that is, $K_i > 0$ and $K_i^{-1} < t$. By Lemma 6.3 the focal points of (M, q) on the line are the points $q + K_i^{-1} v$; such a point lies on the segment from q to $p = q + tv$ if and only if $0 < K_i^{-1} \le t$, and $K_i^{-1} = t$ is excluded because p is not a focal point of (M, q) (q is nondegenerate; Lemma 6.5). Counting multiplicities on both sides gives the claim. □

Milnor writes "eigenvalues … which are $\ge 1/t$"; by the last remark $\ge$ and $>$ agree here. An analogous statement in Part III, the Morse index theorem, will be of fundamental importance: there the straight segment becomes a geodesic, focal points become conjugate points, and L_p becomes the energy of paths.

An interactive look at focal points

The figure shows an oval room: the wall M is an ellipse with semi-axes $a = 10$ m and b (a slider), and the dashed closed curve with four cusps is the evolute, the set of focal points of M. Drag the point p, or use its sliders. The program solves $dL_p/d\theta = 0$ along the wall and draws every normal from p to the wall: solid for minima of L_p (index 0), dashed for maxima (index 1). On each normal a small circle marks the focal point of (M, q), the centre of curvature at its foot q; a normal has index 1 exactly when that circle lies between q and p, as Lemma 6.9 says. Below, the graph of the distance from p along the wall shows the same critical points, and the readout lists the echo delays $2|p - q|/(343\ \mathrm{m/s})$. Drag p across the evolute: a minimum and a maximum approach each other, merge into a degenerate critical point on the evolute, and disappear. The wavefront slider draws the curve at distance s from the wall along the inward normals, where a sound front emitted by the whole wall at once would be after travelling a distance s; its cusps run along the evolute. (The still image shows the listeners at $(3, 2)$, inside the evolute, and $(8, 1)$, outside it, with their distance graphs side by side.)

(In the printed version this figure is a still image; the interactive version is in the web edition.)



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9. Application: echoes in an oval room

The wall and its focal points. Let the wall be the ellipse $M = \{X^2/a^2 + Y^2/b^2 = 1\}$, $a > b > 0$, parametrized by $x(\theta) = (a\cos\theta, b\sin\theta)$; here $k = 1$ and $n = 2$. Put $s(\theta) = |x'(\theta)| = (a^2\sin^2\theta + b^2\cos^2\theta)^{1/2}$ and let $\nu = -(b\cos\theta, a\sin\theta)/s$ be the inward unit normal. Since $x'' = -x$, the second fundamental form in the direction ν is $\nu\cdot x'' = ab/s$, and $g_{11} = s^2$. By Claim B the principal curvature is $K = ab/s^3 > 0$, the curvature of the ellipse, and by Lemma 6.3 the only focal point of $(M, x(\theta))$ is the centre of curvature $x + K^{-1}\nu$. (In the direction $-\nu$ the curvature is $-K$ and the focal point is the same.) Using $a^2 - s^2 = (a^2 - b^2)\cos^2\theta$ and $b^2 - s^2 = -(a^2 - b^2)\sin^2\theta$,

$$x + K^{-1}\nu = \Big(\frac{a^2 - b^2}{a}\cos^3\theta,\ -\frac{a^2 - b^2}{b}\sin^3\theta\Big).$$

So the focal set of M is the *evolute* $(a|X|)^{2/3} + (b|Y|)^{2/3} = (a^2 - b^2)^{2/3}$, a closed curve with cusps at $(\pm(a^2 - b^2)/a, 0)$ and $(0, \pm(a^2 - b^2)/b)$.

The count. The normals from $p = (p_1, p_2)$ are the critical points of $L_p(\theta) = |x(\theta) - p|^2$, and

$$\tfrac{1}{2}L_p'(\theta) = (b^2 - a^2)\sin\theta\cos\theta + ap_1\sin\theta - bp_2\cos\theta.$$

- By Lemma 6.5 and the computation above, L_p is a Morse function if and only if p is off the evolute.
- By Theorem 5.2 of Chapter 3 for the circle ($R_0 = R_1 = 1$), $C_0 = C_1 \geq 1$: off the evolute the number of echoes is even and at least 2.
- By Lemma 6.9, a foot q has index 1 (a local maximum of the distance along the wall) if and only if the centre of curvature at q lies between q and p.
- The number of normals is constant on each component of the complement of the evolute. It is $\#E^{-1}(p)$ (§5), and E is proper: on $N = \{(x(\theta), \tau\nu(\theta))\}$ we have $|E| \geq |\tau| - a$, so preimages of bounded sets are bounded and closed. For a proper smooth map between manifolds of the

same dimension, $\#E^{-1}(p)$ is finite and locally constant on the regular values: see Theorems 5.1 and 5.2 of Chapter 1 of the companion volume, whose proofs use the compactness of the domain only to know that $E^{-1}(p)$ is compact and that E maps closed sets to closed sets; proper maps into $\mathbb{R}^2$ have both properties. The regular values of E are the points off the evolute.

- The complement of the evolute has two components, $\{F < F_0\}$ and $\{F > F_0\}$ with $F = (a|X|)^{2/3} + (b|Y|)^{2/3}$ and $F_0 = (a^2 - b^2)^{2/3}$. Both are connected because $F(\lambda X, \lambda Y) = \lambda^{2/3} F(X, Y)$: the first is star-shaped about 0, and every point of the second can be pushed radially out to a large circle lying in it.
- At $p = 0$, $L'_p = 0$ if and only if $\sin\theta\cos\theta = 0$: four normals. At $p = (R, 0)$ with $R > (a^2 - b^2)/a$, $\frac{1}{2}L'_p = \sin\theta\,(aR - (a^2 - b^2)\cos\theta)$ vanishes only where $\sin\theta = 0$: two normals.

So a listener inside the evolute hears four echoes and one outside it two, as promised in §0. (Apollonius studied the normals from a point to a conic in Book V of the *Conics*; the boundary between two and four normals is implicit there.)

Numbers. Take a room 20 m by 12 m: $a = 10$ m, $b = 6$ m, $a^2 - b^2 = 64$ m^2. The evolute has cusps at $(\pm 6.4, 0)$ and $(0, \pm 10.7)$, so it pokes out through the long walls; the foci are $(\pm 8, 0)$. With the speed of sound 343 m/s (air at 20 °C), an echo from a foot at distance d arrives after $2d/343$ s. Solving $L'_p = 0$ numerically:

listener at	feet of the normals: distance (index)	echo delays
$(0, 0)$, centre	$(0, \pm 6)$: 6 m (0); $(\pm 10, 0)$: 10 m (1)	35.0 ms (two), 58.3 ms (two)
$(5, 0)$	$(7.81, \pm 3.75)$: 4.68 m (0); $(10, 0)$: 5 m (1); $(-10, 0)$: 15 m (1)	27.3 ms (two), 29.2 ms, 87.5 ms
$(8, 0)$, a focus	$(10, 0)$: 2 m (0); $(-10, 0)$: 18 m (1)	11.7 ms, 105.0 ms
$(3, 2)$	3.64 m (0), 7.43 m (0), 7.57 m (1), 13.21 m (1)	21.2, 43.3, 44.2, 77.0 ms

Check one index with Lemma 6.9. For the listener at $(5, 0)$ the radius of curvature of the wall at $(10, 0)$ is $b^2/a = 3.6$ m, so the centre of curvature $(6.4, 0)$ lies between $(10, 0)$ and the listener: index 1. Indeed the 5 m foot is a local maximum of the distance along the wall, flanked by the two feet at 4.68 m, where the radius of curvature, 7.93 m, exceeds the distance: index 0. Walk from $(5, 0)$ toward the end wall and the three near feet slide together; at the cusp $(6.4, 0)$ they merge into one degenerate critical point, and beyond it only $(10, 0)$ is left, now a minimum. The pair of echoes at 27.3 ms and the echo at 29.2 ms fuse into one. The model is idealized: specular reflection, a point listener, and only single reflections.

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10. Summary of what was silently assumed, now made explicit

Where	Milnor says	What is actually needed
p. 32	M "differentiably embedded" in $\mathbb{R}^n$, $k < n$	a smooth submanifold (parametrizations are homeomorphisms onto open subsets); compactness not needed
p. 32	"it is not difficult to show that N is an n-dimensional manifold"	local orthonormal normal frames by Gram–Schmidt; Ψ is a parametrization (Fact A, companion Ch. 6)
p. 33	Sard's theorem, cited	only the smooth case is used; N has a countable basis as a subset of $\mathbb{R}^{2n}$ (companion Ch. 3)
p. 33	the measure-zero set of focal points	a measure-zero set contains no ball, so good p exist arbitrarily close to any point
p. 34	coordinates with $g_{ij}(q) = \delta_{ij}$	a linear change of coordinates; principal curvatures independent of the choice (Claim B)
p. 34	"principal curvatures $K_1, \dots, K_n$"	$K_1, \dots, K_k$
p. 34	fields w_α "along the manifold"	local frames only; the normal bundle can be nontrivial
p. 35	"the rank is equal to the rank of the upper left hand block"	the nullities agree; the ranks differ by $n - k$
p. 35	$(*)$ singular iff $1/t$ is an eigenvalue, with the same multiplicity	$t \neq 0$ (q is never a focal point of (M, q)); symmetry of $(v \cdot \ell_{ij})$
p. 36	"setting $p = x + tv$"	q critical, so $p - q$ is normal; t the distance, v the unit normal; the case $p = q$
p. 36	Corollary 6.7 "on any manifold", by Whitney	Hausdorff with a countable basis; a closed embedding makes M^a compact
p. 36	Application 2: "for any differentiable function f"	f without degenerate critical points; index $(-1)^\lambda$ of grad f; Poincaré–Hopf (companion Ch. 6)
p. 37	Corollary 6.8: "choose some imbedding" h, with bounded image and $h_1 = f$	$h = (f, \beta \circ h')$; derivatives measured in fixed charts covering K

p. 38	eigenvalues "$\geq 1/t$"	$> 1/t$, equality excluded by nondegeneracy; index = number of negative eigenvalues

Checked by machine. Milnor's §6 has been verified in the Lean proof assistant for submanifolds of Euclidean space, closed or not: Theorem 6.1 for smooth maps into a vector space, Corollary 6.2, focal points and their multiplicities, Lemma 6.3 (with principal curvatures as eigenvalues of the shape operator), Assertion 6.4, Lemmas 6.5 and 6.9, Theorem 6.6, and Corollary 6.7 for closed submanifolds. Corollary 6.8 is proved differently, by adding to f a small linear function of a bounded embedding instead of using L_p, and f need not be bounded. Through Mathlib's Whitney embedding, which exists only for compact manifolds, it is checked that every compact manifold carries a Morse function, that Corollary 6.8 holds on it, and that $\chi(M)$ is well defined: $\sum_\lambda (-1)^\lambda C_\lambda$ is the same for every Morse function and equals the index sum of every vector field on any embedded copy, as in Application 2, and the homological Euler characteristic (Chapter 3). Not checked: Corollary 6.7 for non-compact manifolds, Application 1, and the oval room of §9. See the verified edition.